INCH-POUND
MIL-C-17694H(SH)
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(See 6.10)

MILITARY SPECIFICATION

CABLE AND CABLE ASSEMBLIES, SPECIAL PURPOSE, ELECTRICAL, MAGNETIC MINESWEEPING GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for fabrication and test of cables and cable assemblies for magnetic minesweeping.

1.2 Classification. Magnetic minesweeping cables are classified as follows:

P1 Cable - Insulated, nonbuoyant.
P2 Cable - Insulated, nonbuoyant.
P3 Cable - Insulated, nonbuoyant.
A1 Cable - Insulated, buoyant.
A2 Cable - Insulated, buoyant.
A3 Cable - Insulated, buoyant.
S2 Cable - Insulated, single conductor, buoyant.
S3 Cable - Insulated, single conductor, buoyant.
K4 Cable - Noninsulated, single conductor, nonbuoyant.

1.2.1 Standard lengths. Cables are to be in the standard lengths shown in table I (see 6.2). Cables are to be manufactured in sufficient lengths to allow removal of the test samples specified in 4.5.2.1, 4.5.2.2 and 4.5.2.3 without reducing the remaining cable length below the length shown in table I.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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TABLE I. Standard lengths.

Cable	Length (feet)	Tolerance (percent)	
		Plus	Minus
P1	70 1/	5	0
P2	100 1/	5	0
P3	130 1/	5	0
A1	675 2/	2	1
A2	675 2/	2	1
A3	675 2/	2	1
S2	900 2/	2	2
S3	900 2/	2	2
K4	150	2	2

1/ See specification sheets for specific requirements.
2/ Exclusive of flexible copper conductor terminals (pigtails).

1.2.2 Assemblies. Cable assemblies are of the following types as specified (see 6.2):

Type 1 Assembly - One 70 foot P1 section, one 675 foot A1 section, and one 150 foot K4 section.
Type 2 Assembly - One 100 foot P2 section, one 675 foot A2 section, and one 150 foot K4 section.
Type 3 Assembly - One 130 foot P3 section, one 675 foot A3 section, and one 150 foot K4 section.
Type 2 Single assembly (S2 assembly) - One 900 foot S2 section and one 150 foot K4 section.
Type 3 Single assembly (S3 assembly) - One 900 foot S3 section and one 150 foot K4 section.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

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SPECIFICATIONS

FEDERAL

L-P-535 - Plastic Sheet (Sheeting): Plastic Strip: Poly (Vinyl Chloride) and Poly (Vinyl Chloride-Vinyl Acetate), Rigid.

PPP-B-1055 - Barrier Material, Waterproof, Flexible.

MILITARY

MIL-C-17694/1 - Cable and Cable Assemblies, Special Purpose, Electrical Magnetic Minesweeping, Quad Cable and Quad Cable Assemblies.

STANDARDS

FEDERAL

FED-STD-601 - Rubber: Sampling and Testing.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATION

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0981-LP-052-8190 - Cable, Minesweeping Magnetic Type.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

G 23 - Standard Practice for Operating Light-Exposure Apparatus (Carbon-Arc Type) With and Without Water for Exposure of Nonmetallic Materials. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., ATA TRAFFIC Dept., 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.3 Material. Construction materials shall be of the best quality and entirely suitable for the intended application (see 6.3).

3.3.1 Insulation. Synthetic rubber compounds used in the insulation shall be butadiene-styrene copolymer (SBR), chlorosulfonated polyethylene (Hypalon or equal), or a combination thereof. The insulation shall have a resistivity of not less than 3×10^{14} ohms per centimeter (ohm/cm). Insulation compounds shall vulcanize readily to compounds used in the cable jacket and to natural rubber.

3.3.1.1 Reinforcing fabric or braid. Fabric or braid used to reinforce the synthetic rubber in the cable insulation shall be strong, flexible material made of natural or synthetic fibers. It shall be bonded firmly to the synthetic rubber materials. It shall not be damaged by the temperatures used to vulcanize the rubber.

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3.3.2 Jacket. Synthetic rubber compounds used in the jacket shall be polychloroprene or chlorosulfonated polyethylene. Where the insulation material is SBR, the jacket material may be either polychloroprene or chlorosulfonated polyethylene. Where the insulation material is chlorosulfonated polyethylene or a combination of chlorosulfonated polyethylene and SBR, the jacket material shall be chlorosulfonated polyethylene. Jacket material shall be selected to maximize abrasion resistance, tear strength and tensile strength while maintaining resiliency and high flexibility. Jacket materials shall retain the above characteristics when subjected to ozone attack, extreme sunlight, cyclic temperature extremes of minus 37 to 60 degrees Celsius (°C) salt water and oil. Jacket materials shall vulcanize readily to synthetic rubbers used in the insulation and to natural rubber. No reinforcing fabric or braid shall be used in the cable jacket.

3.3.2.1 Toxic products. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.3.3 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using material produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4 General requirements. General requirements shall be as follows:

(a) For A conductors and S cables, high temperature acrylonitrile butadiene styrene (ABS), polycarbonate, or plywood hollow cylinders or cork cylinders shall be used in the cable core.
(b) For P conductors, solid cylinders of high temperature foam polyethylene, polycarbonate, or plywood shall be used in the cable core.
(c) A smaller number of larger size strand conductors may be used provided that the minimum total circular mil area is maintained and flexibility requirements of 3.5.3.7 are met.
(d) The materials shall withstand the maximum operational and repair (localized vulcanizing) temperatures without deformation or permanent change in characteristics. The maximum temperatures for each material shall be determined by thermal analysis of the cable design (see 6.3).
(e) The P/A junction shall be a crimp splice over an aluminum plug as specified in the specification sheets.

3.4.1 Splices. No factory splices shall be allowed in any cable component. The only allowed splice points shall be the terminal cable to P section, the P component to A component, A component to terminal cable, and S cable to terminal cable. Following cable assembly the A/P junction shall be severed and reconnected

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using the splice plugs specified in the specification sheets. These splices shall be crimp connections in accordance with the applicable drawings referenced in the specification sheets and shall withstand 15,000 pounds tension without separation.

3.4.2 Flexibility. Cable assemblies shall have high flexural endurance. Individual strands of conductors shall not break or fray and the internal cylinders shall not crack or separate from the spacers when tested (see 4.7.8). Cable assemblies shall be flexible at all temperatures between minus 37 and 60°C to permit reeling on drums having diameters as follows and as specified in the specification sheets:

S2 single assembly	66 inch diameter drum
S3 single assembly	72 inch diameter drum

3.4.3 Buoyancy. Type P and K4 cables shall be nonbuoyant. Type A and S cables shall have an open end specific gravity not greater than 0.95 as measured in accordance with 4.7.2.

3.4.4 Government furnished equipment. When specified (see 6.2), the Government will furnish the reels specified in 5.1.1.1.

3.5 Cable sheath. The cable sheath shall consist of an inner insulation in accordance with 3.3.1 and an outer jacket in accordance with 3.3.2. The insulation and jacket shall be vulcanized together in such a manner as to produce a single homogeneous sheath about the conductor and shall withstand the ply separation test specified in 4.7.5.3. The sheath of type P cables shall transmit the breaking stresses specified in 3.5.3.8 to the cable grips.

3.5.1 Sheath thickness. The insulation materials used in the sheath shall have a minimum nominal thickness of 120 mils. Jacketing materials shall have a minimum thickness of 50 mils for cable types 1 and 2 and 80 mils for type 3 cable. Where an extruded sheath is used, the combined sheath and insulation thickness shall have a minimum nominal thickness of 170 mils for cable types 1 and 2 and 200 mils for type 3 cable. Extruded sheaths shall be extruded in at least two separate passes and shall contain at least one layer of reinforcing fabric weave or braid meeting the requirements of 3.3.1.1.

3.5.1.1 Dimensional tolerance. The minimum and maximum value of cable sheath thickness with reference to the nominal value shall be as follows:

(a) Average thickness of sheath shall be not less than 90 percent of the nominal value.
(b) Average thickness of sheath shall be not more than 125 percent of the nominal value. Buoyancy requirement shall not be changed.
(c) Any individual thickness shall be not less than 75 percent of the nominal value.
(d) The inside diameters of vulcanizing molds for cable terminations shall be the same as the diameter of the flanges of the respective termination fittings. This requirement is to permit the proper installation of waterproofing boots and sleeves.

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3.5.2 Assembly. The assembly of the sheath shall be in accordance with the requirements of 3.5.2.1 through 3.5.2.5.

3.5.2.1 Terminal cable sheath. The sheath of flexible copper terminal cables used on components of cable assemblies shall be the same as that used on the associated cable component. The molded sheath over terminal lug barrels shall be the same material as is used for the cable jacket.

3.5.2.2 Terminal lug insulation. The insulation of the terminal lugs shall be securely vulcanized to the sheath of the terminal cables and firmly bonded to the terminal barrels and flanges. When tested in accordance with 4.7.6, the adhesion shall be not less than 10 pounds per inch of width.

3.5.2.3 Sheath tapes. Where taped synthetic rubber compounds and insulation are used for the sheath, cable types (P and A sections) shall be assembled with the exposed edges of sheath tapes facing away from the terminal end of the P section. "S" section electrodes shall be attached to the "S" section terminal which faces the exposed edges of sheath tapes.

3.5.2.4 Terminal plugs. The end faces of P cable terminal plugs shall be as specified in the specification sheets.

3.5.2.5 Curing tapes. Curing tapes shall be completely removed from the cable and from the central core of cable groupings.

3.5.3 Physical properties. The cable sheath shall have the physical properties specified in 3.5.3.1 through 3.5.3.8.

3.5.3.1 Tensile strength. The tensile strength of the cable sheath shall be not less than 2000 pounds per square inch (lb/in^2) when tested in accordance with 4.7.5.2.

3.5.3.2 Elongation. The cable sheath shall exhibit an ultimate elongation of not less than 250 percent when tested in accordance with 4.7.5.4.

3.5.3.3 Oxygen and sunlight resistance. The cable sheath shall exhibit no surface checks or cracks when tested in accordance with 4.7.5.5. Following a minimum of 24 hours after this test, the cable sheath shall exhibit not less than 60 percent of the tensile strength and elongation specified in 3.5.3.1 and 3.5.3.2.

3.5.3.4 Dielectric strength. The cable sheath shall withstand without breakdown 2500 volts root mean square (Vrms) 60 hertz (Hz) when tested in accordance with 4.7.1.1. Any cable which fails to meet this requirement may be repaired and retested as specified herein.

3.5.3.5 Insulation resistance. The cable sheath shall exhibit a resistance of not less than 100 megohms when tested in accordance with 4.7.1.2.

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3.5.3.6 Core radial deformation. The deformation of the core cylinders shall not exceed 5 percent of the nominal outside cylinder diameter when tested in accordance with 4.7.7.

3.5.3.7 Flexing. The dc resistance of the cables shall remain constant, conductor strands shall not break, and core cylinders shall not separate from spacers when the cable is tested in accordance with 4.7.8.

3.5.3.8 Minimum breaking strength. The minimum breaking strength with terminals shall be 10,000 pounds for type 1 assemblies and K4 cables and 15,000 pounds for all others.

3.6 Type 1 cable assemblies. Type 1 cable assemblies shall be as specified in the specification sheets and in accordance with the drawings specified in the specification sheets.

3.6.1 Terminals. The P1-A1 junction shall be continuous and marked with a red or other highly visible color band not less than 8 inches in width. The other end of A1 cable shall be terminated as specified in the specification sheets.

3.6.2 Physical and electrical properties. Type 1 cable assemblies shall have the following physical and electrical properties:

(a) Current carrying capacity: 2200 amperes continuous with the buoyant section in sea water at 85 degrees Fahrenheit (°F).
(b) Maximum dc resistance at 25°C: P1/A1 section - 0.028 ohm (excluding the K4 electrode).
(c) DC contact resistance between the P1 and A1 terminal assembly parts shall be not greater than 4 microhms.
(d) Minimum breaking strength with terminals - 10,000 pounds.

3.7 Type 2 cable assemblies. Type 2 cable assemblies shall be as specified in 3.7.1 and 3.7.2 in accordance with the drawings specified in the specification sheets.

3.7.1 Terminals. The P2-A2 junction shall be continuous and marked with a red or other highly visible color band not less than 8 inches in width. The other end of A2 cable shall be terminated as specified in the specification sheets.

3.7.2 Physical and electrical properties. Type 2 cable assemblies shall have the following physical and electrical properties:

(a) Current carrying capacity: 5000 amperes for a cycle of 5-1/2 seconds on and 9-1/2 seconds off with the buoyant section in sea water at 85°F.
(b) Maximum dc resistance at 25°C: P2/A2 section - 0.024 ohm (excluding the K4 electrode).
(c) DC contact resistance between the P2 and A2 terminal assembly parts shall be not greater than 4 microhms.
(d) Minimum breaking strength with terminals - 15,000 pounds.

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3.8 Type 3 cable assemblies. Type 3 cable assemblies shall be as specified in the specification sheets and in accordance with the drawings specified in the specification sheets.

3.8.1 Terminals. The P3-A3 junction shall be continuous and marked with a red or other highly visible color band not less than 8 inches in width. The other end of A3 cable shall be terminated as specified in the specification sheets.

3.8.2 Physical and electrical properties. Type 3 cable assemblies shall have the following physical and electrical properties:

(a) Current carrying capacity: 7500 amperes for a cycle of 5-1/2 seconds on and 9-1/2 seconds off with the buoyant section in sea water at 85°F.
(b) Maximum dc resistance at 25°C: P3/A3 section - 0.0125 ohm (excluding the K4 electrode).
(c) DC contact resistance between the P3 and A3 terminal assembly parts shall be not greater than 4 microhms.
(d) Minimum breaking strength with terminals - 15,000 pounds.

3.9 Type 2 single cable assembly (S2 cable assembly). Type 2 single cable assemblies shall be as specified in 3.9.1 through 3.9.3.

3.9.1 S2 cable. S2 cable shall be a single conductor, self-buoyant, insulated cable suitable for conducting dc from A2 or A1 cable to K4 cable.

3.9.1.1 Construction. S2 cable shall be constructed as shown on the type 1 or type 2 cable assembly drawings specified in the specification sheets.

3.9.1.2 Terminals. Each end of the S2 cable shall be terminated with a flexible connection consisting of:

(a) Parts 563, 559A, and 523, as shown on the type 1 or type 2 cable assembly drawings (see 3.6 and 3.7).
(b) Six feet, 0 inch of 1000 MCM, 61 by 7/.0484 inch MHD copper cable.
(c) Part 562 as shown on the assembly drawings (see 3.6 and 3.7).

3.9.2 Assembly. The S2 cable assembly shall be connected to an A1 or A2 cable as specified in the specification sheets.

3.9.3 Physical and electrical properties. Type S2 cable assemblies shall have the following physical and electrical properties:

(a) Current carrying capacity: 5000 amperes for a cycle of 5-1/2 seconds on and 9-1/2 seconds off with the buoyant section in sea water at 85°F.
(b) Maximum dc resistance at 25°C: S2 section - 0.016 ohm (excluding the K4 electrode).
(c) DC contact resistance between parts 523 and 559A shall be not less than 4 microhms.
(d) Minimum breaking strength with terminals - 15,000 pounds.

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3.10 Type 3 single cable assembly (S3 cable assembly). Type 3 single cable assemblies shall be as specified in 3.10.1 through 3.10.3.

3.10.1 S3 cable. S3 cable shall be a single conductor, self-buoyant, insulated cable suitable for conducting dc from A3 cable to K4 cable.

3.10.1.1 Construction. S3 cables shall be constructed as shown on the type 3 cable assembly drawings specified in the specification sheets.

3.10.1.2 Terminals. Each end of the type S3 cable shall be terminated with a flexible connection consisting of:

(a) Parts 540, 539, and 538, as shown on the type 3 cable assembly drawings (see 3.8).
(b) Seven feet, 4 inches of 2000 MCM, 61 by 19/.0415 inch MHD copper cable.
(c) Part 537 as shown on the assembly drawings (see 3.8).

3.10.2 Assembly. The S3 cable assembly shall be connected to an A3 cable as specified in the specification sheets.

3.10.3 Physical and electrical properties. Type S3 cable assemblies shall have the following physical and electrical properties:

(a) Current carrying capacity: 7500 amperes for a cycle of 5-1/2 seconds on and 9-1/2 seconds off with the buoyant section in sea water at 85°F.
(b) Maximum dc resistance at 25°C: S3 section - 0.007 ohm (excluding the K4 electrode).
(c) DC contact resistance between parts 538 and 539 shall be not less than 4 microhms.
(d) Minimum breaking strength with terminals - 15,000 pounds.

3.11 K4 cable. K4 cable shall be single conductor nonbuoyant cable suitable for establishing an effective current path into or from the sea, using floats to suspend the cable.

3.11.1 Construction. K4 cable shall consist of 61 strands of annealed copper wire, each of 0.1071 inch nominal diameter, concentrically stranded. The cable shall be covered with two layers of fabric tape, each layer applied half-lapped, and No. 9 seine twine braid. Tape and braid of K4 cables shall be rotproofed in accordance with NAVSEA 0981-LP-052-8190.

3.11.2 Terminals. Each end of K4 cable shall be terminated with part 292 as shown in NAVSEA 0981-LP-052-8190.

3.11.3 Physical and electrical properties. K4 cable shall have the following physical and electrical properties:

(a) Maximum dc resistance at 25°C: 0.0160 ohm per 1000 feet.
(b) Minimum breaking strength with terminals - 10,000 pounds.

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3.12 Cable identification. The information listed in 3.12.1 shall be clearly molded into the insulation over the terminal barrels on both ends of the cables. In addition, the serial numbers of the cables shall be stamped into the ends of the terminals using steel stamps with 3/4-inch characters.

3.12.1 Identification information. Identification information shall be as follows:

(a) Cable designation (for example, PQ2, AQ2, and S2).
(b) Manufacturer.
(c) Year of manufacture.
(d) Serial number.
(e) National stock number.

3.12.2 Serial numbers. Serial numbers will be furnished by the command or agency concerned.

3.12.3 Reels. Cable reels shall meet the requirements of 3.12.3.1 through 3.12.3.3.

3.12.3.1 Identification plates. Both flanges of each reel of cable shall have securely affixed thereto identification plates made by heat sealing between two approximately 0.020 inch thick clear, transparent, vinyl sheets, in accordance with L-P-535, a paper printed with the marking information required (see 5.2.1). Identification plates of other types providing equivalent legibility and durability may be used when specifically approved by the command or agency concerned.

3.12.3.2 Color series. Each reel of cable shall be identified with a keyed series of colors to indicate the year of manufacture. This identification shall consist of a painted stripe approximately 2 inches wide and colored as specified herein for the year of manufacture. The cycle of colors shall be repeated every fifth year.

Year of manufacture	Identifying color
1989	Blue
1990	White
1991	Red
1992	Green
1993	Orange
1994	Blue

3.12.3.2.1 Color stripe. The colored stripe shall be applied circumferentially over the lagging (see 5.1.1.4) and midway between the flanges. The stripe shall consist of one coat, commercial quality, outside paint of appropriate color. In addition to the stripe, both flanges of each reel shall be stenciled with 4-inch high figures to show the year of manufacture.

3.12.3.3 Caution notice. The caution "DO NOT USE FORK LIFT" shall be painted conspicuously on each flange of each reel.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material (see 6.3).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. For any cable design not previously tested, a sample test cable of each cable assembly type shall be provided to the Navy for first article testing. Sample test cables shall be 18 feet long including 4 foot pigtails at each end. Pigtails shall have end lugs installed in such a manner that with the sample cable laid out in a relaxed state the surface of both terminal lugs lie in the same plane. The sample test cable shall contain a resistance temperature device (RTD) installed at each point in the cable where maximum temperature is expected. Electrical leads from each RTD shall extend out of the cable. Leads shall be 10 feet long. Leads shall be insulated and shall be vulcanized to the cable sheath material in a manner to ensure the cable assembly is water-tight (see 6.6). The RTD shall be installed in a manner that does not change the general geometry of the cable. First article inspection will consist of the examinations specified in 4.6 and the tests specified in 4.7.

4.3.1 Pigtail connections. Pigtails shall be connected to the sample cable as specified for the completed cable assemblies. The cable assembly or each cable when not included in an assembly, will be tested as shown in table II.

4.4 Quality conformance inspection. Quality conformance shall consist of the examinations and tests specified in table II of the samples chosen in accordance with 4.5.

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4.4.1 Lot acceptance and routine tests. The cable assembly or each cable when not included in an assembly shall be tested as shown in table II. Lot acceptance and routine tests shall be as specified in table II. A lot is defined in 4.5.1. Routine test shall be performed on all cable assemblies, or cable procured.

TABLE II. Required performance test.

Test	Paragraph	Lot acceptance					Routine acceptance				
		ASM	P	A	S	K4	ASM	P	A	S	K4
Buoyancy	4.7.2			X	X						
Compression	4.7.4			X	X						
Contact resistance 1/	4.7.1.3						X	X	X	X	
Dielectric	4.7.1.1						X	X	X	X	
Examination	4.6						X	X	X	X	X
Flexing	4.7.8		X	X	X						
Inspection of packaging	4.8				X			X	X		
Insulation resistance	4.7.1.2						X	X	X	X	
Radial deformation	4.7.7			X	X						
Resistance	4.7.9						X	X	X	X	
Sheath thickness	4.7.5.1						X	X	X	X	
Sheath tensile strength	4.7.5.2	X	X	X	X						
Sheath separation test	4.7.5.3		X	X	X						
Sheath elongation	4.7.5.4	X	X	X	X						

See footnote at end of table.

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TABLE II. Required performance test - Continued.

Test	Paragraph	Lot acceptance					Routine acceptance				
		ASM	P	A	S	K4	ASM	P	A	S	K4
Sheath oxygen and sunlight resistance	4.7.5.5	X	X	X	X						
Tensile strength	4.7.3			X	X						
Terminal lug insulation adhesion	4.7.6	X	X	X	X						

1/ These values to be measured during construction of cable.

4.5 Sampling. Samples for test and inspection shall be obtained as follows:

4.5.1 Lot. Unless otherwise specified (see 6.2), an inspection lot shall consist of not more than 25 cables of one type.

4.5.2 Cable samples. One cable shall be selected at random from the first six production cable assemblies of a lot (or from the complete lot if the quantity is six or less) and the test samples specified in 4.5.2.1, 4.5.2.2 or 4.5.2.3 taken from it. The cable assembly shall then be reterminated. Removal of the test sample shall not reduce the length of the cable assembly to less than the tolerance specified herein. Completed cable assemblies or each cable (except electrodes), if not included in an assembly, shall be examined in accordance with 4.6 and tested in accordance with 4.7.

4.5.2.1 P cable sections. A 20-foot plus or minus 5 percent sample of P cable shall be cut from the nonbuoyant end of type P/A cable assemblies selected in accordance with 4.5.2. The components specified in the specification sheets shall be separated, cut to a length of 5 feet, terminated in the same manner as the existing terminations, and subjected to the test specified in 4.7.3.1. Another component specified in the specification sheets shall be subjected to the test specified in 4.7.8.

4.5.2.2 A cable sections. A 20-foot plus or minus 5 percent sample of A cable, along with the associated flexible connections and terminal fittings, shall be cut from the buoyant end of type P/A cable assemblies selected in accordance with 4.5.2. The conductors specified in the specification sheets shall be cut to a length of 5 feet plus or minus 10 percent and the cut ends terminated with the same type of flexible connections and terminal fittings as already on the sample, except that the duplex connector shall remain separated. These conductors shall

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then be subjected to the test specified in 4.7.3.1. Three feet plus or minus 10 percent shall be cut from the remaining part of the specified components and subjected to the tests specified in 4.7.2. Another specified component shall be subjected to the test specified in 4.7.8. Another specified conductor and its terminal lug shall be subjected to the tests specified in 4.7.4 and 4.7.6.

4.5.2.3 S cable sections. A 48-foot plus or minus 5 percent sample of S cable, along with the associated flexible terminal cables, connectors, and lugs, shall be cut from one end of type S cable assemblies selected in accordance with 4.5.2. Three feet plus or minus 5 percent of the buoyant conductor shall be cut from the severed end of the sample and subjected to the tests specified in 4.7.2 and 4.7.4. Five feet plus or minus 5 percent shall be cut off from the terminated end of the sample. This 5-foot section shall be terminated on the out end in the same manner as the original cable assembly and subjected to the tensile strength test specified in 4.7.3.2. This remaining 40-foot section shall be subjected to the test specified in 4.7.8.

4.5.2.4 Configuration changes. A 10-foot test sample of each cable containing a configuration change from the guidance drawings contained in NAVSEA 0981-LP-052-8190 shall be submitted for testing.

4.5.2.5 Retesting. In the event of failure of any of the samples to pass the tests specified in 4.7.2, 4.7.3, or 4.7.4, one additional cable assembly shall be selected at random from the same lot and duplicate samples taken. All of the conductor samples shall be subjected to the tests failed. Failure of any one of the samples from the second cable selected shall be grounds for rejection of the lot.

4.5.2.6 Sheath. One 2-inch wide circumferential sample of cable sheath shall be taken from each end (except P-A connection) of every P and A cable conductor. Two 2-inch wide circumferential samples of cable sheath shall be taken from each end of every S cable. The samples shall be tagged with the serial number of the cable from which removed. (These samples are from the sheath removed from terminal installation.) These samples shall be used for the sheath thickness test specified in 4.7.5.1. The samples obtained from the first six production cables of a lot shall be used for the tests specified in 4.7.5.2, 4.7.5.3, 4.7.5.4, and 4.7.5.5.

4.6 Examination. Production cable shall be examined to determine conformance to this specification as to dimensions, materials, workmanship, and general construction. Total resistance measurements shall be measured carefully, using accepted standard procedures. Additional examinations shall be performed as specified in the specification sheets.

4.7 Tests. Tests on cables and assemblies shall be performed as follows:

4.7.1 Electrical tests. Electrical test shall be performed as specified in 4.7.1.1 through 4.7.1.3.

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4.7.1.1 Dielectric. A dielectric test on each completed cable assembly or each cable (except electrodes), when not included in an assembly, shall be performed by immersing the cable in water for 4 hours, after which it shall be subjected to a potential of 2,500 Vrms (60 Hz) for 5 consecutive minutes between the conductor and ground. If the test should be interrupted, the total duration shall be increased 20 percent of the specified duration for each interruption. Any cable which fails to meet this requirement may be repaired and again tested.

4.7.1.2 Insulation resistance. At completion of the dielectric test specified in 4.7.1.1 and while the cable is still in the water, the insulation resistance shall be measured using a 500 volt megohmmeter. Insulation resistance corrected to a temperature of 60°F shall be not less than 100 megohms as measured between the installed terminals and ground. For quad assemblies, two measurements are required: one for paralleled long leg conductors and one for the paralleled short leg conductors.

4.7.1.3 Contact resistance. A dc contact resistance test shall be made on each pressed connection of each complete cable to determine conformance to the resistance values specified in 3.6.2, 3.7.2, 3.8.2, 3.9.3 and 3.10.3.

4.7.2 Buoyancy test. A 2- to 3-foot sample of each type of buoyant cable selected from the first lot under each contract shall be floated in a tank of fresh water at room temperature. Lead weights shall be applied along the length until the whole sample just sinks. The open end specific gravity is then calculated from the following:

$$\text{Specific gravity} = \frac{\text{Wt. of cable in air}}{(\text{Wt. of cable in air}) + (\text{Wt. of lead in air})} \times \frac{10.4}{11.4}$$

4.7.3 Tensile strength. Tensile strength test shall be as specified in 4.7.3.1 and 4.7.3.2.

4.7.3.1 A cables. The samples selected in accordance with 4.5.2.2 shall have the newly terminated ends connected by means of an equalizing bridle or yoke and shall be tested as follows: A1 cable shall have a tensile load of 5,000 pounds and A2 and A3 cables shall have a tensile load of 7,500 pounds applied between bridle and yoke and end fitting for 4 hours. There shall be no permanent damage or deformation of the cable after test.

4.7.3.2 S cables. The samples of S cables (see 4.5.2.3) shall have a tensile load of 15,000 pounds applied between end fittings for 4 hours. There shall be no permanent damage or deformation of the cable after test.

4.7.4 Compression test. A sample of each buoyant cable (see 4.5.2.2 and 4.5.2.3) shall be subjected to the following compressive force values for a period of 4 hours, applied between two flat planes 6 inches in length:

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Type of cable	Compressive force (pounds)
A1	2,000
A2	2,500
S2	3,000
A3	3,000
S3	3,000

The test shall result in no permanent deformation or damage to the cable.

4.7.5 Sheath tests. The cable sheath shall be subjected to tests as specified in 4.7.5.1 through 4.7.5.5.

4.7.5.1 Thickness. At least five separate measurements shall be made on each of the sheath samples taken in accordance with 4.5.2.6. The average of all the measurements taken of all samples from each cable assembly shall be considered the average thickness. Where the materials used in the insulation and jacket differ, the requirements of this paragraph shall apply to the insulation and jacket individually. Tolerances specified in 3.5.1.1 shall apply.

4.7.5.2 Tensile strength. The jacket material of five sheath samples from each lot (see 4.5.2.6) shall be tested for tensile strength using method 4111 of FED-STD-601. One specimen from each sample shall be tested. The average tensile strength of the specimens shall be considered the tensile strength of the sheath and shall be not less than 2000 lb/in^2.

4.7.5.3 Sheath separation. Five sheath samples from each lot (see 4.5.2.6) shall be tested for strength of bond between adjacent sheaths or between successive layers of extruded sheath and between the insulation and jacket using method 4221 of FED-STD-601 except that the force shall be applied in such a manner as to tend to separate the individual layers of insulation and jacket along their mating faces. One specimen from each sample shall be tested. Bond between individual sheath jacket and insulation, or between either of these and an intervening reinforcing tape, braid, or serving shall be not less than 10 pounds per inch of width.

4.7.5.3.1 Thermal cycling and torque test. A 2-foot length of component shall be subjected to 10 thermal cycles of heating to predicted operating temperature followed by forced cooling to room temperature. A 3-inch long section of sheath on this sample shall be isolated by removing a minimum of 4 inches of sheath on either side. A torque shall be applied to this isolated section in the direction to twist the sheath section around the cable. The sheath shall withstand a torque of at least 60 foot pounds without signs of separation from the underlying conductors.

4.7.5.4 Elongation. Five sheath samples from each lot (see 4.5.2.6) shall be tested for elongation of jacket material using method 4121 of FED-STD-601. The jacket samples shall be obtained by slicing the jacket material from the sheath samples. One specimen from each sample shall be tested. Ultimate elongation shall be not less than 250 percent.

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4.7.5.5 Oxygen and sunlight resistance. Jacket material from a minimum of five sheath samples from each lot (see 4.5.2.6) shall undergo an accelerated aging procedure consisting of the exposures specified in methods 7111 and 7311 of FED-STD-601. Twenty-four hours after accelerated aging is complete, the tensile strength and elongation shall meet requirements specified in 3.5.3.3. Surface checks and cracks shall not be visible to the naked eye. One specimen from each sample shall be tested. As an alternative to test method 7311 for sunlight resistance, 100 hour exposure in type D test equipment in accordance with ASTM G 23 may be used employing a cycle of approximately 100 minutes of arc light exposure followed by approximately 18 minutes of exposure to arc light and water spray.

4.7.6 Terminal lug insulation adhesion. A test specimen shall be prepared from one of each type of terminal lugs selected in accordance with 4.5.2.1 and 4.5.2.3. The adhesion of the insulation to the lug barrels shall be tested by method 8031 of FED-STD-601. The sheath shall be reduced to a thickness of approximately 3/16-inch over the undistorted area of the lug during preparation of the specimen. Adhesion shall be not less than 10 pounds per inch of width.

4.7.7 Radial deformation. Twelve core cylinders shall be subjected to a deformation test with the load applied perpendicular to the axis of the cylinder. The deformation at loads tabulated below shall not exceed 5 percent of the nominal outside cylinder diameter.

Cable	Nominal outside diameter of cylinder (inches)	Load (pounds)
A1	1.60	800
A2	1.75	1400
A3	2.50	2000
S2	2.50	2000
S3	3.25	2600

4.7.8 Flexing. Samples of component cable assemblies, as specified in 4.5.2.1, 4.5.2.2 and 4.5.2.3, shall be wrapped around the diameter mandrel specified below, for 1.5 turns. A 100-pound weight shall be attached to each free end of the cable by means of truncated Kellum grips for distributing load circumferentially around the cable jacket. The mandrel shall be rotated so that each end of the cable is lowered and raised a distance of 3 feet. Raising the cable end 3 feet then lowering it 3 feet to return to its original position is defined as 1 cycle. A total of 100 cycles shall be performed. The dc resistance shall remain constant within plus or minus 1 percent of the average dc resistance throughout test.

Cable assembly type	Mandrel diameter (inches)
A1/P1	24
A2/P2	26
A3/P3	36
S2	66
S3	72

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4.7.9 Resistance. The final dc resistance of each cable assembly shall be measured prior to shipment by applying 10 amperes dc through the cable component and recording the millivolt drop across the cable length. Resistance shall be in accordance with 3.6.2, 3.7.2, 3.8.2, 3.9.3, and 3.10.3 when corrected to 25°C.

4.8 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7.)

5.1 Preservation and packing. Preservation and packing shall be level A as specified herein.

5.1.1 Preservation.

5.1.1.1 Cable assemblies. S cables with K4 electrodes attached shall be wound on separate reusable steel reels manufactured in accordance with NAVSEA 0981-LP-052-8190. Steel reels for type P/A cable assemblies with K4 electrodes attached, shall be manufactured in accordance with the drawings specified in the specification sheets. P/A cable assemblies, with one K4 electrode attached, shall be wound with the terminals on top and cables shall be wound with an attached K4 electrode on the bottom. Both ends of the cable shall be securely fastened to the reel with rot resistant rope.

5.1.1.2 Electrodes. K4 electrodes, when acquired separately, shall be wound on sturdy wooden shipping reels. Two K4 electrodes shall be wound on each reel.

5.1.1.3 P cables. P cables, when acquired separately, shall be wound on sturdy shipping reels of appropriate diameter.

5.1.1.4 Lagging. The cable wound on reels shall be covered with flexible waterproofed wrapping paper conforming to PPP-B-1055 and secured to prevent unwrapping during handling and shipment. Reels shall be completely enclosed with wooden lagging with boards meeting each other. Laggings shall be 2-inch commercial lumber and shall be strapped with at least three 1-inch wide steel straps, except that reels containing K4 electrodes need have only two straps. Straps shall be stapled at intervals in accordance with the contractor's commercial practice. A minimum clearance of one cable diameter shall be provided between the inner face of the lagging and the waterproof wrapping material. When wooden reels are used, the lagging shall be cut even with the outside of the reel flanges and nailed with cement coated steel nails. Lagging of steel reels shall be in accordance with the applicable steel reel drawing (see 5.1.1.1).

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5.1.2 Packing. Cable assemblies or separately procured cables preserved as specified in 5.1 shall not require overpacking. The method of packing or loading of reels shall be accomplished in a manner which will ensure acceptance by carrier and will afford protection against physical or mechanical damage during shipment from the supply source to the using activity. The method of loading shall conform to the National Motor Freight Traffic Association Rules and the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required (see 6.2), each reel shall be marked in accordance with MIL-STD-129.

5.2.1 Special marking. Each reel shall be marked as follows on the identification plates as specified in 3.12.3.1:

(a) Cable type.
(b) Cable serial number.
(c) Contract number.
(d) Manufacturer's name and destination.
(e) Federal stock number.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cables described in this specification are intended for use during magnetic minesweeping.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Type of cable or cable assembly required (see 1.2 and 1.2.2).
(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When first article inspection is required (see 3.2).
(e) Whether the Government will furnish the reels specified in 5.1.1.1 (see 3.4.4).
(f) Lot quantity if special (see 4.5.1).
(g) Special marking requirements (see 5.2.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific

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acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.3	DI-MISC-80678	Certification/ Data Report	-----
3.4	DI-S-30591	Data and/or analysis summary	-----
4.1.1	DI-MISC-80653	Reports, test	-----

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.2), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Drawings. Drawings referenced herein should be in accordance with the latest edition of NAVSEA 0981-LP-052-8190.

6.6 Resistance temperature device (RTD). A chart relating to the ohmic resistance of the RTD to the measured temperature should be prepared.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.4.4, if required.

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6.9 Subject term (key word) listing.

Sheath, vulcanized
Terminal

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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